

Youssef Elbehiry

youssef.elbehiry99@gmail.com • +49 177 596 0441 • Duisburg, Germany • linkedin.com/in/youssef-elbehiry • Born: 20 Sep 1999

Summary

B.Sc. Mechanical Engineering graduate (UDE, Thesis Grade: 1.0, July 2026), enrolled in M.Sc. Mechanical Engineering at UDE from Winter Semester 2026/27. Hands-on experience in thermohydraulic systems and safety-relevant rotating equipment at a 500 MW combined-cycle power station. Strong foundation in thermodynamics, fluid dynamics and MATLAB/Simulink modelling. Skilled in mechanical design, 3D CAD, FEA simulation, GD&T and Lean process improvement.

Work Experience

- Digital Trust & Safety Analyst — Telus Digital, Essen | Part-time 20 h/week**11/2023 – Present
 - Evaluate 200+ digital content items per shift against safety policies; analyse risk patterns and proactively flag potential threats; maintain quality targets alongside full-time engineering studies and thesis.
- Industrial Internship — MEGA for Construction & Industries | Cairo West Power Station**06–09/2025500 MW combined-cycle plant, Egypt | 13-week internship
 - Overhauled thermohydraulic systems (centrifugal pumps, heat exchangers, boiler feed); assessed safety-relevant components; ensured quality and safety compliance; documented findings in technical reports.
 - Applied vibration spectrum analysis for predictive maintenance; conducted hydraulic performance testing to qualify restored systems against design specifications.

Education

- B.Sc. Mechanical Engineering — University of Duisburg-Essen, Germany**10/2021 – 07/2026GPA: 2.0 | Thesis Grade: 1.0 | Graduated: July 2026 | M.Sc. Mechanical Engineering — enrolled from WS 2026/27**Thesis:** "Analysis of Influences on Roll Force, Torque and Forward Slip in Hot Flat Rolling of Steel by Fast Simulation" (03.07.2026)**Modules:** Thermodynamics, Fluid Dynamics, System Dynamics & Control (MATLAB/Simulink), Engineering Dynamics (FEM), Machine Design
- B.Sc. Mechanical Engineering — Mansoura University, Egypt**09/2017 – 07/2021GPA: 1.3**Modules:** Manufacturing Processes, Machine Design, CAD, Engineering Drawing, Thermodynamics

Skills

- CAD & Design:** SolidWorks – Advanced (CSWA Certified), AutoCAD – Intermediate, CATIA V5 – Intermediate, NX Siemens – Intermediate
- Simulation & Analysis:** MATLAB/Simulink – Intermediate, ANSYS – Intermediate, FEM – Intermediate
- Measurement & Testing:** Thermohydraulic performance testing; vibration spectrum analysis; dimensional inspection; GD&T
- Lean & Quality:** Lean Six Sigma White Belt
- IT & Data:** MS Office – Proficient, Python – Basic, C – Basic, SAP – Basic, Power BI – Basic
- Soft Skills:** Technical Documentation, Team Leadership, Structured Analytical Thinking, Cross-cultural Communication

Technical Projects

- Bachelor Thesis (Grade: 1.0): Hot Flat Rolling Process Analysis & Simulation — UDE**2026
 - Evaluated six analytical rolling theories; validated against 52 measured passes on C55 steel across five parametric studies (pass reduction, roll-gap geometry, interstand tensions, friction, multistand rolling).
- HHO Electrolyzer | Team Leader — Mansoura University**2020
 - Designed electrolyzer in SolidWorks; selected AISI 316L electrodes for corrosion resistance; conducted electrochemical efficiency measurements and validated hydrogen production against design targets. Led 4-person team to prototype.
- Dual-Axis Solar Tracking System | Team Leader — Mansoura University**2019
 - Designed a dual-axis solar tracker using Arduino and 4 LDR sensors; modelled mechanical frame in SolidWorks. Led a team of 6 through design, servo actuation integration and prototype delivery.

Certifications

- CSWA – Certified SolidWorks Associate (Mechanical Design) |** Dassault Systèmes | 2020
- Lean Six Sigma White Belt |** Educate 360 | 2026
- Introduction to Composite Manufacturing |** LinkedIn Learning | 2026
- MATLAB Onramp |** MathWorks | 2026
- A Hands-on Introduction to Engineering Simulations |** Cornell University (CornellIX) | 2026
- Introduction to Geometric Dimensioning and Tolerancing |** LinkedIn Learning | 2026
- Simulink Onramp |** MathWorks | 2026
- German Language Certificate – B1 |** Goethe-Institut | 2021

Languages

German: B2 • English: C1 • Arabic: Native